

Assignment 1

To capture live network packets using Wireshark and identify basic network protocols such as TCP, UDP, IP, and DNS.

Solution:

The image shows a Wireshark network packet capture. The top pane displays a list of captured packets. The bottom pane shows the details of the selected packet (No. 1, Time 0.000000).

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	BrocadeCommu_90:80:78	Broadcast	ARP	60	Who has 172.16.77.61? Tell 172.16.64.1
2	0.000229	fe80::17:26c3:b88e::	ff02::1:ff47:97ec	ICMPv6	86	Neighbor Solicitation for fe80::877:758a:41a7:97ec from 22:31:6a:15:4c:01
3	0.001293	172.16.75.82	224.0.0.251	MDNS	110	Standard query 0x0000 TXT Aaditya's MacBook Air_companion-link._tcp.local, "QU" question
4	0.001293	fe80::cfb:e951:7143::	ff02::1:ff09:4067	ICMPv6	86	Neighbor Solicitation for fe80::1ca3:924a:84d9:4067 from 6a:fb:25:d2:4a:eb
5	0.001293	fe80::1473:7b51:e69::	ff02::fb	MDNS	130	Standard query 0x0000 TXT Aaditya's MacBook Air_companion-link._tcp.local, "QU" question
6	0.001293	172.16.84.231	224.0.0.251	MDNS	106	Standard query 0x0000 TXT HAC's MacBook Air_companion-link._tcp.local, "QM" question
7	0.001293	fe80::188b:c255:ccc::	ff02::fb	MDNS	126	Standard query 0x0000 TXT HAC's MacBook Air_companion-link._tcp.local, "QM" question
8	0.002635	172.16.77.146	224.0.0.251	MDNS	1469	Standard query 0x0000 PTR _airplay._tcp.local, "QM" question PTR Namit's MacBook Air_airplay._tcp.local PTR Aryaman's MacBook Air
9	0.003078	fe80::1c46:1930:b77::	ff02::fb	MDNS	1489	Standard query 0x0000 PTR _airplay._tcp.local, "QM" question PTR Namit's MacBook Air_airplay._tcp.local PTR Aryaman's MacBook Air
10	0.003078	fe80::41b:cb42:8b54::	ff02::fb	MDNS	127	Standard query 0x3fc0 PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
11	0.003078	fe80::cabd:49bb:f06::	ff02::fb	MDNS	127	Standard query 0x76e5 PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
12	0.003699	fe80::24c6:d3e7:171::	ff02::fb	MDNS	127	Standard query 0x5420 PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
13	0.003699	fe80::cabd:49bb:f06::	ff02::fb	MDNS	127	Standard query 0x0f40 PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
14	0.004114	fe80::5d77:4b72:627::	ff02::fb	MDNS	127	Standard query 0x1a3b PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
15	0.004114	fe80::a006:b313:920::	ff02::fb	MDNS	127	Standard query 0x35f4 PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
16	0.004114	fe80::5d77:4b72:627::	ff02::fb	MDNS	127	Standard query 0xf66d PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
17	0.004114	fe80::41b:cb42:8b54::	ff02::fb	MDNS	127	Standard query 0xbdf1 PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
18	0.004114	fe80::6367:14e8:cbe::	ff02::fb	MDNS	127	Standard query 0x4291 PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
19	0.005263	fe80::24c6:d3e7:171::	ff02::fb	MDNS	127	Standard query 0x862c PTR YajatPahuja's MacBook Air_airplay._tcp.local, "QM" question
20	0.005263	172.16.77.146	224.0.0.251	MDNS	1469	Standard query 0x0000 PTR Avaneendra's MacBook Air_airplay._tcp.local PTR Akshat's MacBook Air (4)_airplay._tcp.local PTR Jess

Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF{...}

Ethernet II, Src: BrocadeCommu_90:80:78 (74:8e:f8:90:80:78), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request)

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: BrocadeCommu_90:80:78 (74:8e:f8:90:80:78)

Sender IP address: 172.16.64.1

Target MAC address: 00:00:00:00:00:00 (00:00:00:00:00:00)

Target IP address: 172.16.77.61

Target MAC address (arp.dst.hw_mac), 6 bytes

Packets: 70668 - Dropped: 0 (0.0%)

Profile: Default